

Let's start by examining the empty spaces in the target rectangle and the provided polyominoes and their variations. We need to fit the Tetromino I, Tetromino L, and Tetromino T into the empty spaces without overlapping any existing polyominoes or going outside the rectangle.

The target rectangle with 12 empty squares looks like this:



We have two variations for Tetromino I (■), which can either be placed vertically or horizontally. Given the configuration of the empty spaces, the only way to place the Tetromino I is horizontally at the top row, as the vertical placement would not fit anywhere.

After placing Tetromino I (■), the target rectangle looks like this:



Next, we have two variations for Tetromino L (■). We can see that Variation 3 of Tetromino L can fit in the bottom left corner, while Variation 7 would not fit anywhere without overlapping or going outside the rectangle.

After placing Tetromino L (■), the target rectangle looks like this:

...



...

Now, we have two variations for Tetromino T (). We need to fit this into the remaining empty spaces. Variation 2 of Tetromino T cannot fit because it would require two empty spaces next to each other vertically, which we do not have. However, Variation 3 of Tetromino T fits perfectly into the remaining space.

After placing Tetromino T (), the target rectangle looks like this:

...



...

The correct variation of Tetromino T that fits into the empty squares is:

B. 3

(b) Response of Visual Tiling - Page 2

Figure 14: Responses of Visual Tiling

Let's visualize the grid with the items you've described:

```
+---+---+---+
| S | B | M |
+---+---+---+
| P | C | V |
+---+---+---+
| T | W | C |
+---+---+---+
```

Where:

- T = Cassette player
- W = Wool
- C = Conch
- V = Moving van
- C = Confectionery store
- P = Pot pie
- S = Siamang
- B = Black-and-white colobus
- M = Minivan

You start at the bottom-left corner where the cassette player is located (T). Now, let's follow your steps:

1. Go right by one step (to Wool - W):

```
+---+---+---+
| S | B | M |
+---+---+---+
| P | C | V |
+---+---+---+
| T | *W* | C |
+---+---+---+
```

2. Go right by one step (to Conch - C):

```
+---+---+---+
| S | B | M |
+---+---+---+
| P | C | V |
+---+---+---+
```